# MECH 454

# Project 1

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# Topic 1 - Hybrid Powertrains

An automotive powertrain refers to the components that generate and transmit power to the wheels of a vehicle [cook]. Meaning that it would comprise components such as: the engine, the transmission, the driveshaft, the axle, the differential, and could even contain more components if someone is trying to recreate and realistic model. These components help transform and transmit the chemical energy of fuel into mechanical energy, all the way down to the wheels. All of these components are characterized using metrics such as: torque, power, fuel consumption, efficiency, etc [book]. While some are more common than others thanks to the car community, their origin stems from physics and engineering terminology; more specifically metrics in thermodynamics. When generalizing, most of the previously mentioned components focus on the "transmission" of power, which often result in some sort of energy lost in the process. However, the engine is the only component that you could say is "generating" power due to the transformation of chemical energy into mechanical energy through combustion. Therefore, it's often regarded as the heart and soul of vehicles as other parts mostly work around it. Parts like the transmission, use a series of gears and gear ratios to maximize the powerband of the engine. The axle is responsible for transmitting the power coming out of the transmission to the front, back, or all 4 wheels [book].

The following figure illustrates the components of a powertrain in a simple manner, where the flow of power is clearly visible; where every change in step implies some sort of loss.

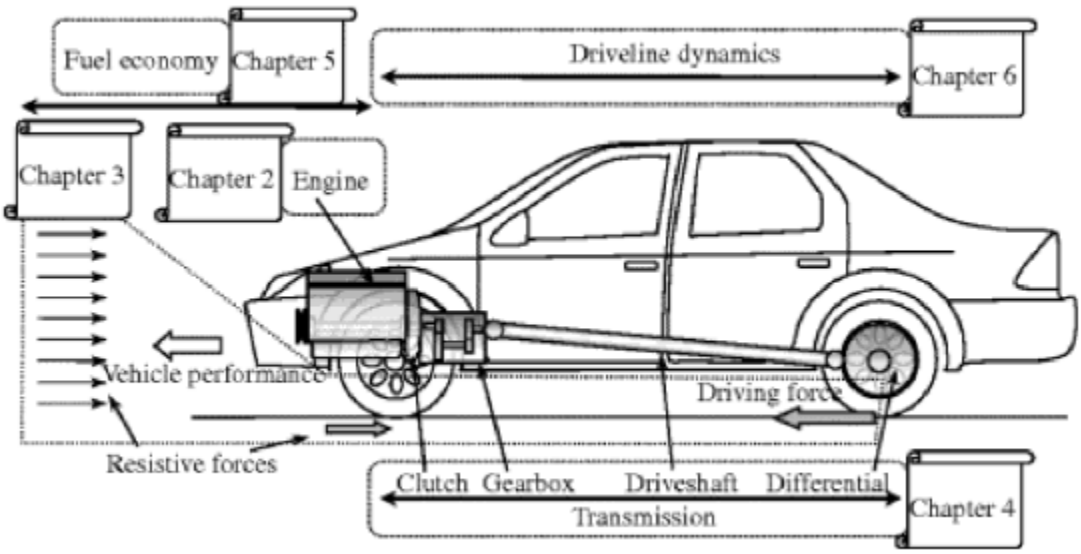

Figure 1: Flow of power throughout the powertrain [book]

A hybrid powertrain, or more also referred as hybrid electric vehicle (HEV), uses an Internal Combustion Engine (ICE) as well as an electric motor powered by a chemical battery [mehrdad, 18]. Although variations of this exist, they generally use two or more different modes of power generation that work together to make them more efficient and give them better fuel economy [mehrdad, 18]. Figure 2 illustrates an example of the power flow of an HEV, where the direction in which the powertrain propels as well as charges the battery; improving the efficiency and performance compared to a conventional ICE engine which only utilises one.

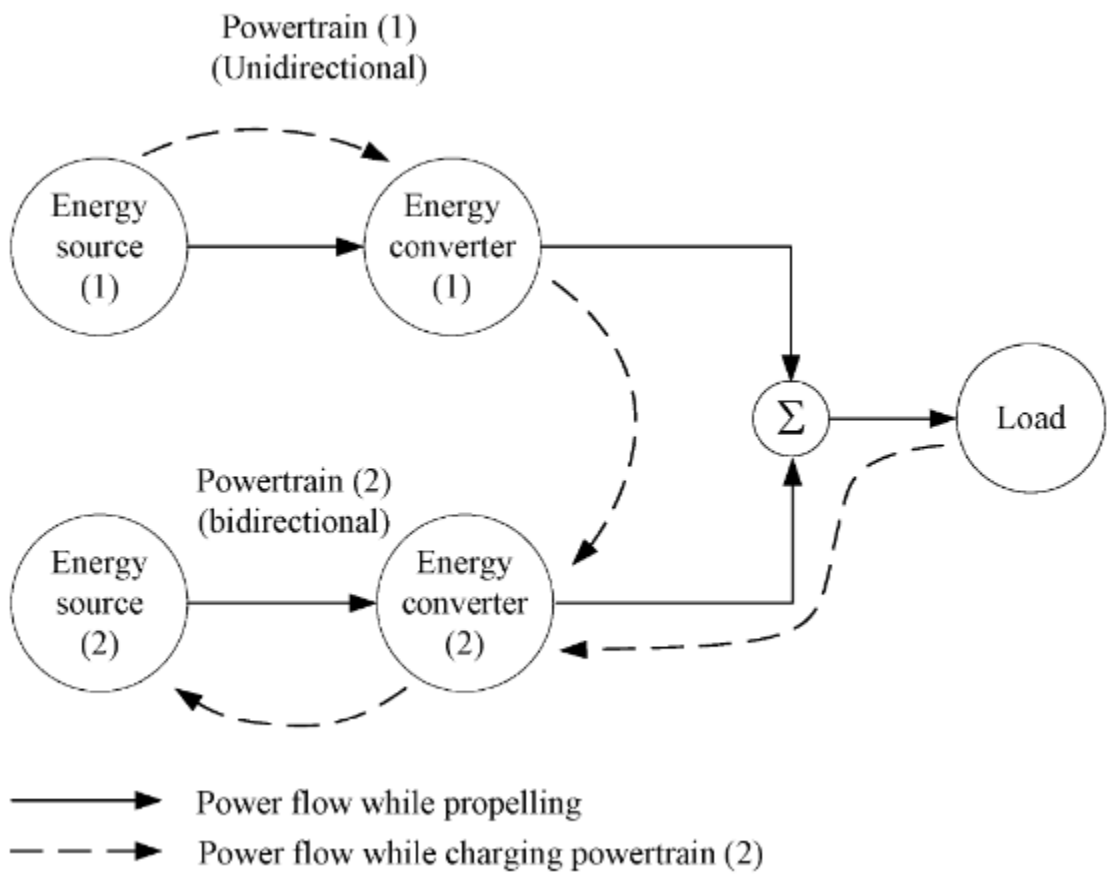


Figure 2: Illustration of hybrid powertrain diagram

The load is considered to vary at random, due to it being associated to the inertia of the car, slopes, acceleration, deceleration, and many more factors. Therefore the load is dynamic, and the same can be said for other types of drivetrains. However, there is another dotted line

between the load and the electric powertrain, which represents the change in inertia recharging the battery or also known as regenerative braking [book]. There are various architectures or topologies of hybrid powertrains, where some common examples are: Series hybrid, Parallel hybrid, Series-Parallel, and Combination HEV [kamil, 18].

A Series hybrid is an architecture that uses the ICE to drive a generator, which in turn charges a battery, which then drives an electric motor that are connected to the wheels. This means that there is no direct mechanical coupling between the gas engine and the wheels (load). It also uses ultracapacitors to increase energy storage and in turn increase efficiency. The disconnect between the two powertrains allows the powertrain to have smaller engines, carbon footprints, longer lifespan [kamil, 18].

The Parallel hybrid is the architecture shown in *Figure 2*, where both powertrains are connected to the transmission. This allows for the ICE to operate constantly at maximum efficiency [kamil]. Here, the power flows from the ICE or the battery/electric motor to the wheels; and in the case of the electric drivetrain it can be recharged by the ICE engine [kamil]. This architecture is regarded as being simpler than other hybrid drivetrains due to a simpler electrical system. This topology also allows for the drivetrain to have higher efficiency at highway speeds, due to the load being shared amongst the two drivetrains. Its main disadvantage is that high voltages are needed for optimal efficiency [kamil].

The third topology is a combination of parallel and series HEV, which combine both previous architectures into one. This design allows it to not fully rely on a single drivetrain, allowing the system to switch between any combination to optimize fuel efficiency or power delivery [kamil, 18]. This architecture flexibility is what allowed this topology to become the most common one for passenger vehicles [5]. A series of planetary gears connect to the ICE and the two electric machines with the transmission serving as an intermediary between both powertrains, as seen in Figure 3.

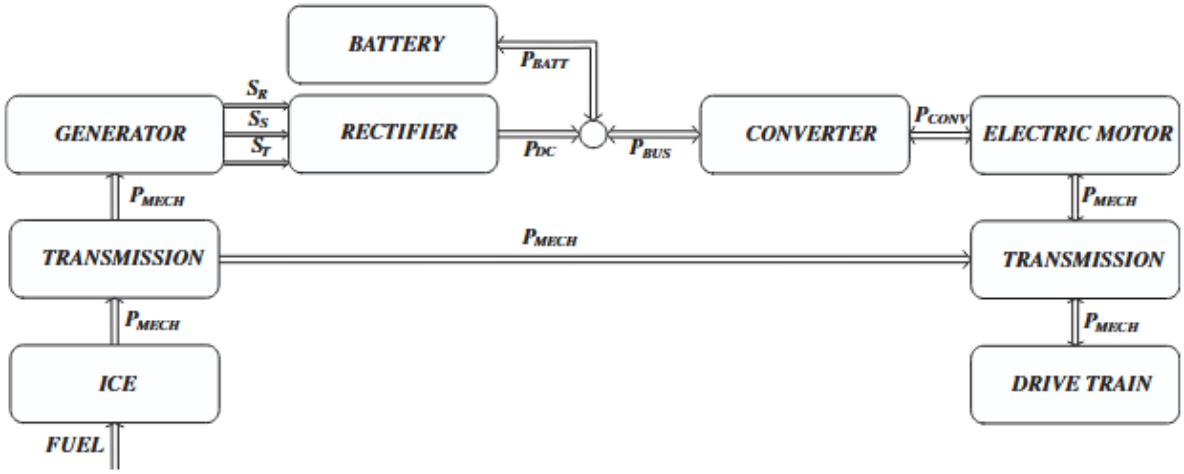


Figure 3: Illustration of combination of parallel and series HEV diagram

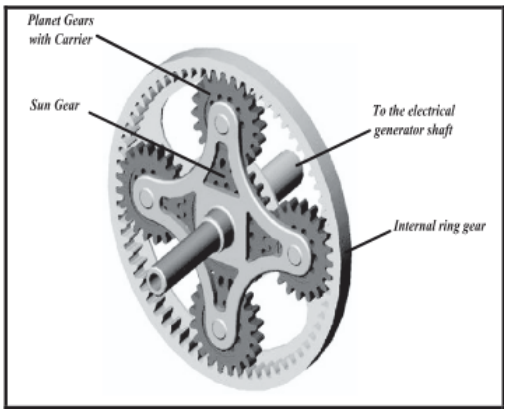


Figure 4: Planetary gears linking ICE, generator and Electric motor

In addition, the speed of the engine can be regulated by changing the speed of the generator, hence the importance of the planetary gear in the whole system.Due to the alternator being able to control the speed of the engine, this allows it to operate in its most efficient state at every speed; making it one of the most efficients [kamil]. The main downside is due to complex algorithms and control systems required to allow the system to work properly and efficiently.

# Topic 2 - Future of ICE

The vehicle market, more specifically cars and vans, is still mostly dominated by the Internal Combustion Engine making up around 80% of all vehicles on the road [6]. With ICEs dominating the vehicle market, they're extremely susceptible to fluctuations in oil and gas prices as seen recently with global conflicts. In addition, the energy market is shifting away from oil and gas, which is why hybrid technologies have been becoming larger players in the market and why governments are enforcing some zero-emission vehicle sales regulations [7].

The role of ICE is still relevant in today's transportation sector, and is especially clear when comparing the alternatives. Although alternative fuels, hydrogen cars, and other alternatives exist, they still pale in comparison when compared to the recent rise in popularity of EV's and the rise of electric in the transportation industry. Although it might seem like EV's and hybrid are everywhere, they still only make up around 15% of sales of light-duty automobiles in 2023 [6]. This illustrates how combustion engines are still very relevant in today's world; where that 15% of sales includes plug-in hybrids and battery electric vehicles, meaning some part of those sales still have ICEs. Governments such as Canada's have implemented incentives to try to shift people towards EV's, however sources indicate that "sluggish" EV sales and mineral rarity might prove to be a challenge in their timeline [9, 10]. Combustion engines provide high energy density, efficiency, driving range, technology maturity, and more, that are all benefits that companies and individuals will consider when purchasing a vehicle.

First, when analyzing the energy density of the main technologies in transportation, the winners and losers become clear. The 3 fuels that we'll compare are Hydrogen, Gasoline, and Diesel; where they have: 143 MJ/kg, 46.4 MJ/kg, and 45.4 MJ/kg respectively [11]. If we compare these Li-ion batteries in their current state using *Figure 5* [8] and converting to MJ/kg, we get around ~0.72 MJ/kg. For a more fair comparison, more recent battery technologies could be used for this comparison, however it becomes clear the energy density (energy per mass unit) is substantially higher for gasoline and diesel fuels versus batteries. The definite winner is by far Hydrogen vehicles with the best energy density out of the mentioned fuels and batteries, however this is a topic for another paper. This means that unless battery technologies advances drastically, the combustion engine remains hard to fully replace with EVs since their lower energy density leaves a performance gap regarding driving range. When adding Hybrid vehicles to the equation, they have around 120 to 180 Wh/kg which is around 0.43 to 0.648 MJ/kg [12].

Although these numbers are similar in energy density to EVs, they have increased efficiency as well as lower overall weight.

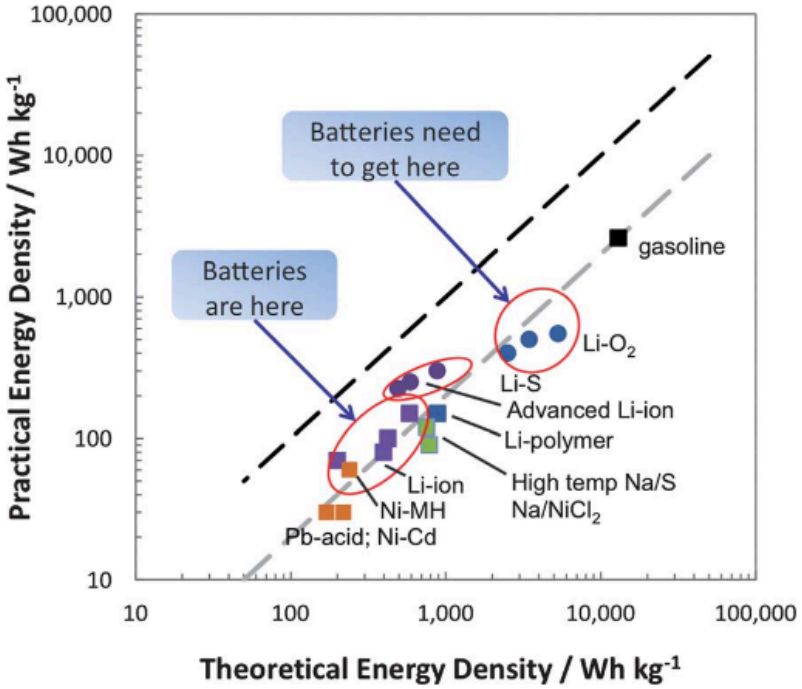


Figure 5: Plot of Practical Energy Density/Wh kg-1 vs Theoretical Energy Density/Wh kg-1 [8]

Second, is the driving range of vehicles as well as efficiency. As covered in the last paragraph, the energy density of hybrid engines at first glance appears as low as EVs with Li-ion batteries. However, their advantage comes with efficiency, where vehicles using ICEs have an efficiency of 12%-30% vs over 77% in EVs [13]. Hybrid powertrains achieve higher efficiencies than regular combustion engines due to them combining both powertrains to obtain a more efficient outcome as previously discussed. A study from *Natural Resources Canada* compares the fuel economy of: ICEV (Internal Combustion Engine Vehicles), HEV (Hybrid-Electric Vehicles), and BEV (Battery Electric Vehicles). There's an interactive figure that allows us to compare the fuel economy based on vehicle class using equivalent Litres (Le). It becomes clear how combustion engines generally have lower efficiencies than EVs, but its clear that the hybrid vehicles sit at the middle point between both sides [13]. However, the gap is noticeably bigger when comparing city consumption vs highway consumption due to the constant need to accelerate

and decelerate; where BEVs have significantly more technology to recuperate lost energy, as well as having better output efficiency [13]. This constant change in acceleration makes ICEs consume more gasoline due to their powertrain efficiency being very low. However, having a higher weight due to energy density requires the BEVs to haul a larger overall weight. In other words, even if it appears as if BEVs are the clear winner due to fuel consumption, their weight reduces their real usable driving range due to “stop-and-go” [19]. Looking at vehicle manufacturers’ claimed range, it becomes clear how Hybrid powertrains try to obtain the most range out of both ICE and BEV. For example: Tesla’s Model Y base model has a claimed range of 463 km, Toyota claims their base model Prius has a total range of 890 km, and Ford’s base model Bronco’s can easily be calculated based on their given specs as seen in *Equation 1* [14, 15, 16, 17]. Although an argument can be made that EVs are relatively close in range to the more common ICE, we notice that HEVs have almost double their range. This gap in *real* range is due to the HEV powertrains such as the combination of parallel/series hybrid powertrain using the ICE and electric motors/batteries in a way to ensure optimal power flow delivery. This makes HEVs a rather good choice for long range travel since it requires less refilling.

$$Estimated\ Range\ =\ Fuel\ Efficiency\ \ *\ \ Fuel\ Tank\ Volume$$
$$544km\ =\ 8.50\ km/L\ \ *\ \ 64\ L$$

Equation 1: Estimated range for Ford Bronco base model

Third, the infrastructure poses a challenge for EV manufacturers to overcome to gain a larger footprint in the vehicle market. As gasoline and diesel engines have been widely adopted for decades, it has become the default choice for many vehicle users. This has allowed the technology to achieve a certain level of maturity that becomes noticeable when observing the infrastructure in place for both technologies [18, 19]. The difference can be noticed when looking for an EV charger versus when looking for a gas station, a gas station can be found with little to no effort whereas for an EV charger are not as common. In many cases, it even requires users to use apps and websites to find the nearest gas station or plan road trips based on availability of charging stations along the way. Although this might seem like a small detail, it indirectly favours some Hybrid topologies over others, as some architectures are more convenient for longer ranges compared to others.

Fourth, the role ICE lies in Hybrid vehicles. In the wake of the transition from ICEs to EVs, Hybrid vehicles serve as a gradual transition which use both power sources combined to overcome some limitations of EVs such as energy density, while striving towards an electrified transport sector [19]. Some EV limitations were previously covered such as infrastructure and technology maturity, but by using HEVs, users are enabling these technologies to mature while also reducing emissions [19]. HEVs users contribute to the development and spreading of infrastructure such as charger; as demands rises for them, companies and governments will fill in with the supply [19]. Hybrid powertrains are especially promising when looking at industries such as shipping and trucking, where data suggests that EV truck sales in the United States was 1200, which amounts to only 0.1% of all truck sales [20]. For reference, earlier it was mentioned that amongst passenger vehicles, around 15% of sales in 2023 were PHEV and EVs [6]. This highlights a gap in the market where EVs haven't fully established themselves. Hybrid powertrains could help fill this market gap, whilst reducing tailpipe emissions, whilst also improving the usable range; which would be an economic factor that trucking companies consider.

In conclusion, Internal Combustion Engines are no longer the only powertrain available in the market, with EVs gaining a foothold due to its efficiency and low carbon footprint. However, the transition to EVs proves to be a challenge, as lots of its infrastructure is still underdeveloped and some technology still needs to mature. Hybrid powertrains and architectures serve as a stepping stone that allows its EV technologies to mature, while still giving the consumer the added benefit of a smaller carbon footprint, increased efficiency, smaller barrier of entry, and many more. HEVs get the best of both the ICE and electric motors/batteries, at an affordable price point while reducing the tailpipe emissions of many passengers and company fleets.

# Topic 3 - Idealized Cycle calculations

The code can all be found in the Q1 file in the following repo:

https://github.com/Rickyy26/MECH-454-Project-1

# Deliverable 1:

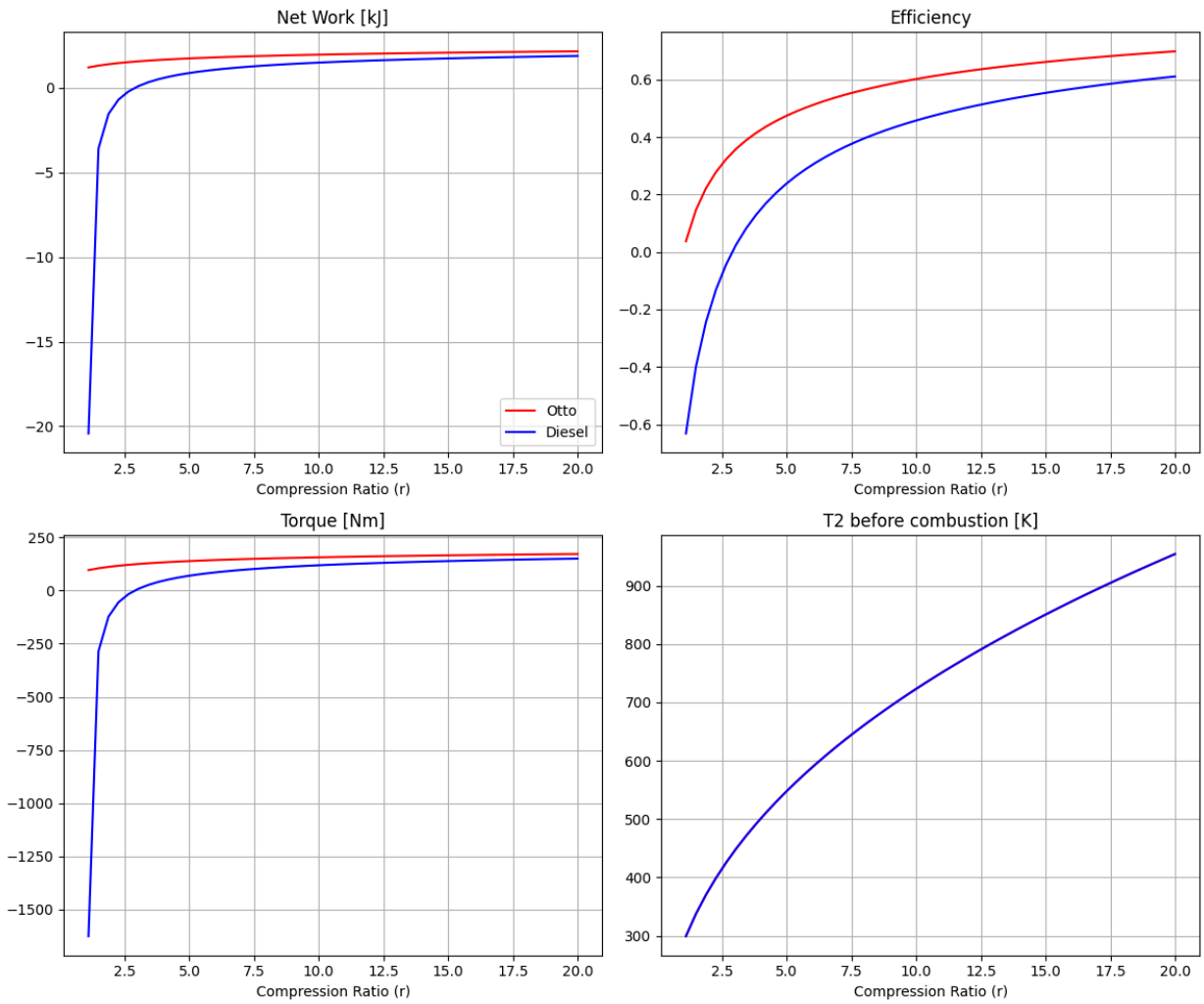


Figure 3: Resulting graphs for Q1

# Deliverable 2:

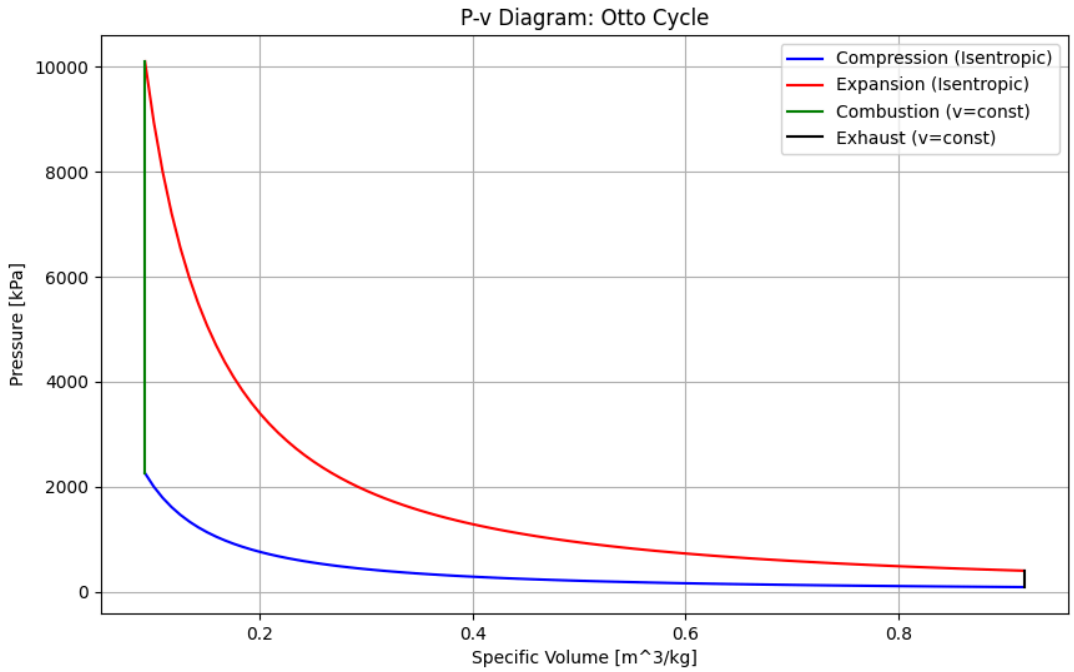


Figure 4: P-V Diagram for Otto Cycle for Question 2

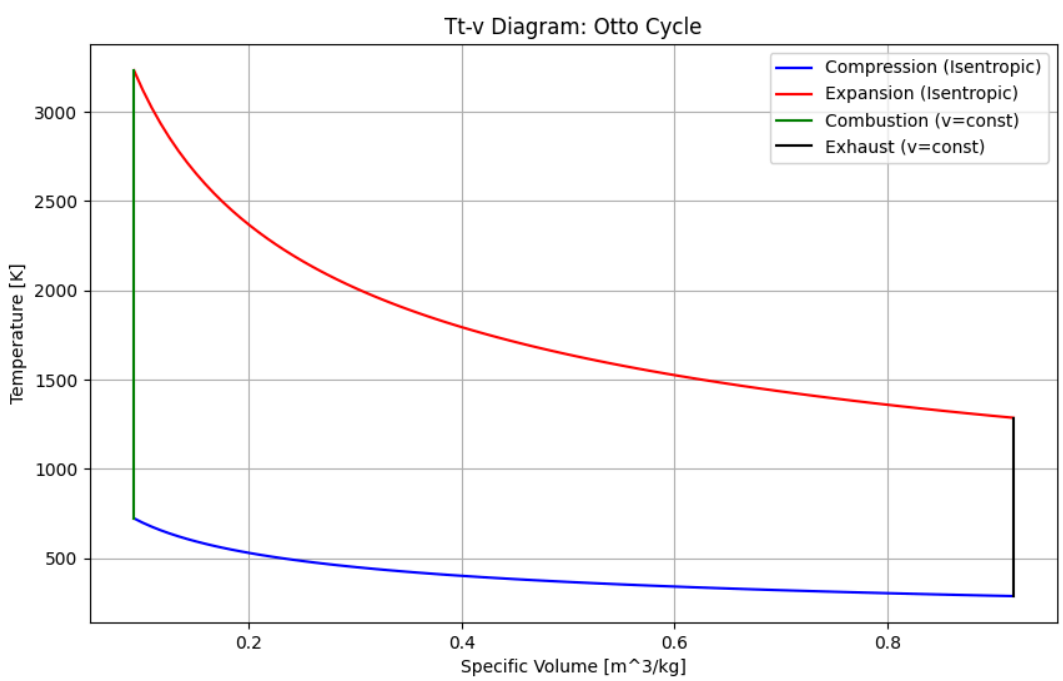

Figure 5: T-V Diagram for Otto Cycle for Question 2

Figure 6:

## Deliverable 3:

When comparing the results obtained from Question 3 to Question 1, we obtain a percent difference of $5.56 * 10^{-6}$, meaning it's extremely close to the original value from Question 1. The output from the code for question 3 is:

```
 Question 3
Analytical T2 (Q1): 723.4233 K
Numberical T2 (Q3): 723.4233 K
Difference: 5.56e-06 %

Case  | Int W [kJ]   | State W [kJ] | Error %
-------------------------------------------------
1     | -500.4244    | -500.4244    | 0.00e+00
2     | -556.0272    | -556.0272    | 0.00e+00
3     | -500.4244    | -500.4244    | 0.00e+00
4     | -566.9574    | -566.9574    | 4.01e-14
5     | -626.6160    | -626.6160    | 0.00e+00
6     | -564.3900    | -564.3900    | 0.00e+00

 State      | Temp [K]   | Pres [kPa]   | Vol [m^3/kg]
-------------------------------------------------------
st1        | 288.00     | 90.00        | 0.9184
st2        | 723.42     | 2260.70      | 0.0918
st3        | 3232.13    | 10100.42     | 0.0918
st4        | 1286.74    | 402.10       | 0.9184

--- Results for OTTO ---
Efficiency: 60.19%
Net Work: 1966.11 kJ
Compression Work: 566.96 kJ
```

Expansion Work: 2533.07 kJ

Torque: 156458.33 Nm

# Deliverable 4:

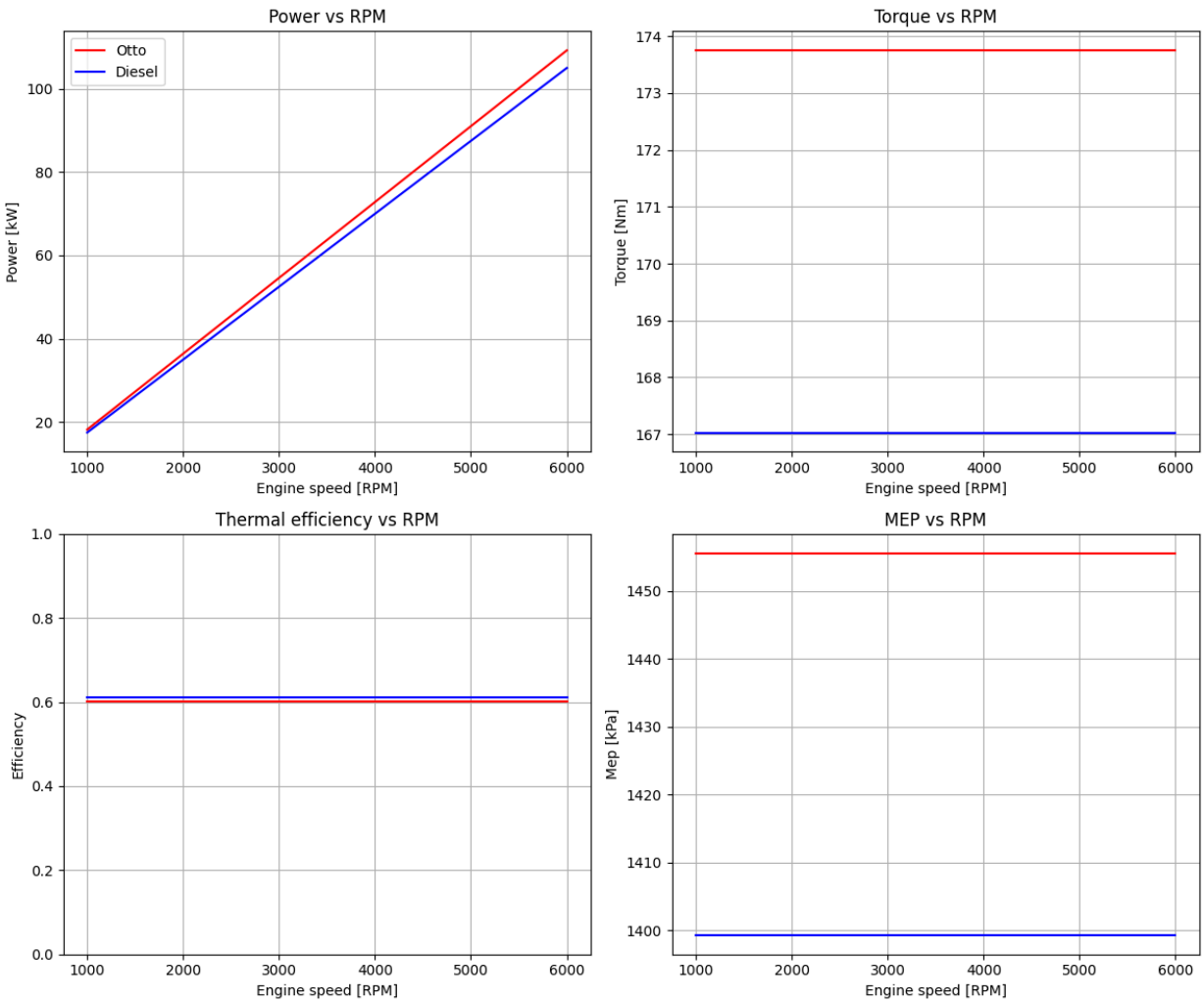


Figure 7: Resulting graphs for Q4

In the 4 generated plots, we notice the following differences between the cycles:

- Power vs RPM: we notice there's a perfect linear increase in both models. This is most likely caused by the fact that they're both ideal models of engines. No losses are being considered that would normally reduce performance the higher we go in RPM's.
- Torque, efficiency, and MEP: these 3 graphs result in a flat lines, which displays their constant power output. This is due to the ideal models as previously mentioned, which

ignores factors such as friction losses, transmission losses, combustion efficiency, and even load on the engine.

- Otto and diesel: The otto appears to be better than the diesel engine in every simulation except for efficiency, where the diesel engine beats out the otto. However, the torque and MEP should be higher for a the diesel engine rather than the otto. This could be due to the fact that in the simulation parameters, we used the same air-fuel mix ratio. Normally, these two engines don't share the same AF mixture, with otto normally having around 14.7 as a ration [purdue]. This means that the otto engine is running rich, but since this is an ideal cycle that doesn't account for combustion efficiency, we get higher performance numbers.
- Compared to real engines: We don't notice common power curve shapes which vary; as was mentioned previously, the losses normally reducing performance aren't being considered in this simulation. Mechanical friction, varying efficiency based on engine temperature, combustion duration, etc. These are all factors that are being neglected which greatly influence the results.

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